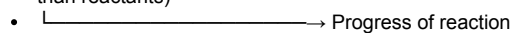
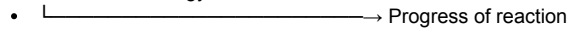


C5 Chemical Energetics — Comprehensive Study Guide — Cheat Sheet

Key Terms

Exothermic reaction	A reaction that transfers thermal energy to the surroundings, causing an increase in temperature
Endothermic reaction	A reaction that takes in thermal energy from the surroundings, causing a decrease in temperature
Enthalpy (H)	The total thermal energy (heat) content of a chemical system
Enthalpy change (ΔH)	The transfer of thermal energy during a reaction; also called heat of reaction
Energy level diagram	A diagram showing the energy levels of reactants and products, indicating whether a reaction is exothermic or endothermic
Bond energy (E)	The energy required to break one particular type of covalent bond; the same energy is released when that bond forms
Activation energy (E_a)	The minimum amount of energy that colliding particles must possess for a reaction to occur
Bond breaking	The process of pulling atoms apart by overcoming attractive forces; always endothermic
Bond making	The process of atoms coming together to form new bonds; always exothermic
System	The chemicals involved in the reaction
Surroundings	Everything outside the system — the container, air, water, thermometer

Key Equations

- 1. Energy flows out of the system into the surroundings → exothermic
- 2. Energy flows into the system from the surroundings → endothermic
- Combustion (burning fuels) — burning methane, wood, petrol: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$
- Neutralisation — acid + alkali → salt + water
- Displacement reactions — adding zinc to copper(II) sulfate solution: $\text{Zn}(\text{s}) + \text{CuSO}_4(\text{aq}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu}(\text{s})$
- Photosynthesis — $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (plants absorb light energy)
- Thermal decomposition — breaking down compounds with heat, e.g., $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- Arrow pointing downwards → exothermic (products have LESS energy than reactants)
- Arrow pointing upwards → endothermic (products have MORE energy than reactants)
-  → Progress of reaction
- $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ — $\Delta\text{H} = -728 \text{ kJ/mol}$
- $2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightarrow 2\text{MgO}(\text{s})$
- This reaction gives off heat and light → exothermic → ΔH is negative → products are drawn below reactants.
- 1. Bonds in the reactants must be BROKEN — this requires energy input → endothermic process
- 2. New bonds in the products are MADE — this releases energy → exothermic process
- Net result: energy is released → exothermic overall
-  → Progress of reaction
- ΔH negative → exothermic → energy released → products lower than reactants
- ΔH positive → endothermic → energy absorbed → products higher than reactants
- Methane burns in oxygen according to: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$, $\Delta\text{H} = -728 \text{ kJ/mol}$. Explain what the negative sign of ΔH tells you

about this reaction.

Exam Focus

1. Define exothermic and endothermic correctly: always refer to "transfer of thermal energy to/from the surroundings" and the "increase/decrease in temperature of the surroundings." Do NOT just say "gives out/takes in heat."

Key Facts

- C5.01 Exothermic and endothermic reactions — Understanding that all reactions involve energy changes, classifying them as exothermic or endothermic, reading and drawing energy level diagrams, the concept of enthalpy change (ΔH), bond breaking vs bond making, and the role of activation energy.
- The system = the reacting chemicals themselves
- The surroundings = everything else — the beaker, the water, the air, the thermometer, your hand
- Key point for exams: You MUST describe exothermic and endothermic in terms of energy transfer to/from the surroundings, not just "gives out heat" or "takes in heat".
- Memory aid: "EXothermic = energy EXits the reaction"
- Common examples of exothermic reactions:
 - Combustion (burning fuels) — burning methane, wood, petrol: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$
 - Neutralisation — acid + alkali → salt + water
 - Oxidation reactions — rusting of iron, respiration
 - Displacement reactions — adding zinc to copper(II) sulfate solution: $\text{Zn}(\text{s}) + \text{CuSO}_4(\text{aq}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu}(\text{s})$
- Everyday exothermic processes:
 - Memory aid: "ENdothermic = energy ENters the reaction"
 - Common examples of endothermic reactions:
 - Photosynthesis — $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ (plants absorb light energy)
 - Thermal decomposition — breaking down compounds with heat, e.g., $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
 - Dissolving ammonium nitrate in water — used in cold packs
 - Citric acid + sodium hydrogencarbonate — temperature drops noticeably
- Everyday endothermic processes:
 - Exam tip: The question may describe a reaction where the temperature of the reaction mixture increases — this means the reaction is transferring heat TO the mixture (exothermic), causing it to warm up. The surroundings of the specific chemicals gain this thermal energy.
- How to read an energy level diagram:
 - Horizontal axis = Progress of reaction (from reactants on the left to products on the right)
 - Vertical axis = Energy (in kJ or kJ/mol)
 - Worked Example — Methane combustion:
 - Exam tip: When asked to state whether a reaction is exothermic or endothermic from ΔH , look at the sign: negative = exothermic, positive = endothermic. You must always include the sign when writing ΔH values.
 - Step 1: Write a word equation or balanced symbol equation for the reaction

Quick Q&A

Question	Answer
Which type of reaction takes in heat from its surroundings?	An endothermic reaction takes in thermal energy from the surroundings, causing a decrease in the temperature of the surroundings. ---